



# Finite population estimation under generalized linear model assistance

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## ABSTRACT

Finite population estimation is the overall goal of sample surveys. When information regarding auxiliary variables are available, one may take advantage of general regression estimators (GREG) to improve sample estimates precision. GREG estimators may be derived when the relationship between interest and auxiliary variables is represented by a normal linear model. However, in some cases, such as when estimating class frequencies or counting processes means, Bernoulli or Poisson models are more suitable than linear normal ones. This paper focuses on building regression type estimators under a model-assisted approach, for the general case in which the relationship between interest and auxiliary variables may be suitably described by a generalized linear model. The finite population distribution of the variable of interest is viewed as if generated by a member of the exponential family, which includes Bernoulli, Poisson, gamma and inverse Gaussian distributions, among others. The resulting estimator is a generalized linear model regression estimator (GEREG). Its general form and basic statistical properties are presented and studied analytically and empirically, using Monte Carlo simulation experiments. Three applications are presented in which the GEREG estimator shows better performance than the GREG one.

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## 1. Introduction

Unbiased estimation of finite population means, totals and percentages, based on sample surveys using probability sampling, can be accomplished by the Horvitz–Thompson estimator. When information regarding auxiliary variables is available, however, one may take advantage of general regression estimators (GREG) to improve sample estimates' precision. Such GREG estimators may be derived under a model assisted estimation approach (see, for example, [Särndal et al., 2003](#)). They also can be derived when the relationship between interest and auxiliary variables may be represented by a normal linear model, under a prediction approach for finite populations. Several authors, such as [Bolfarine and Zacks \(1992\)](#), [Isaki and Fuller \(1982\)](#), [Wright \(1983\)](#) and [Fuller \(2002\)](#), have been working on the subject of regression estimators. Nevertheless, since the efficiency of the GREG methodology lies on the ability of the formulated regression model to describe the concomitant behavior of the interest variable and the auxiliary ones, sometimes it is necessary to consider using models to deal appropriately with cases where the scale measurement of the interest variable is not continuous or it is continuous but highly skewed. Considering broader classes of models is also needed for cases in which the relationship between the interest variable and the auxiliary ones is clearly not linear. [Lehtonen and Veijanen \(1998\)](#), for example, considered deriving a Logistic General Regression Estimator (LGREG) using a multinomial response model to describe such relationship, for estimating class frequencies. [Breidt and Opsomer \(2000\)](#) focused on a nonparametric regression modeling assisted approach,

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**Table 1**  
Principal distributions belonging to exponential family.

| Distribution | $b(\theta)$          | $\theta$              | $\phi$            | $V(\mu)$     |
|--------------|----------------------|-----------------------|-------------------|--------------|
| Normal       | $\theta^2/2$         | $\mu$                 | $1/\sigma^2$      | 1            |
| Poisson      | $e^\theta$           | $\log \mu$            | 1                 | $\mu$        |
| Bernoulli    | $\log(1 + e^\theta)$ | $\log\{\mu/(1-\mu)\}$ | 1                 | $\mu(1-\mu)$ |
| Gamma        | $-\log(-\theta)$     | $-1/\mu$              | $1/(\text{CV})^2$ | $\mu^2$      |
| Inverse G.   | $-\sqrt{-2\theta}$   | $-1/2\mu^2$           | $\phi$            | $\mu^3$      |

proposing and studying the properties of local polynomial regression estimators. [Estevao and Särndal \(2004\)](#) studied several applications of domain estimation using calibration. [Duchesne \(2003\)](#) compared the efficiency of an LGREG and a GREG based on a normal linear model using Monte Carlo simulation. [Lehtonen et al. \(2003\)](#) compared the performance of different estimators, including those assisted by the logistic regression model. [Lehtonen et al. \(2005\)](#) studied the importance of model specification for estimating the total of a polytomous interest variable for a number of large or small domains. [Myrskylä \(2007\)](#) investigated variance estimation for the LGREG estimator for domains. [Li \(2008\)](#) introduced the Box–Cox transformation into the generalized regression estimator, which can be especially suitable to deal with highly skewed continuous and positive interest variables, where normal response models may not be appropriate. Using a model-assisted estimation approach, this paper focuses on proposing regression estimators for the general case in which the relationship between interest and auxiliary variables may be appropriately described by a generalized linear model. The finite population distribution of the interest variable is viewed as if generated by a member of the exponential family distribution, which includes normal (even with variance heterogeneity), Bernoulli, Poisson, gamma and inverse Gaussian distributions. The resulting estimator is a generalized linear model regression estimator (GEREG). The GEREG setup is flexible enough to accommodate classical estimators, such as the ratio estimator, as particular cases. Therefore, it is expected that a GEREG estimator would perform at least as better as a GREG estimator on those situations where linear models would not fit the data as well as generalized linear models. This paper introduces the GEREG estimator in its general form and their statistical properties, taking into account a broad class of sampling designs. We consider the usual inference setup for finite population, based on a sample  $S$ , of size  $n$ , selected from a population, of size  $N$  ( $n < N$ ), according to a probability sample with first and second-order inclusion probabilities given by  $\pi_k = P(k \in S)$  and  $\pi_{kl} = P(k, l \in S)$ , respectively.  $y_k$  is defined to be the value of the interest variable, measured at element  $k \in U$ , for direct element sampling schemes. When considering a one-stage cluster design, we regard  $U$  as the set of clusters listed on a sampling frame, and  $y_k$  as the aggregated value of the variable of interest, for cluster  $k \in U$ . In any case, we denote by  $\mathbf{x}_k = (x_{k1}, \dots, x_{kj})^T$  the auxiliary information vector for the element (or cluster)  $k \in U$ . The paper is organized in four sections: Section 1 is an introduction; Section 2 is a brief review of generalized linear models for finite population; in Section 3 we present the GEREG estimator and its main statistical properties. We also present a simulation study that illustrates the performance of the GEREG estimator. Section 4 presents three applications of the proposed estimator.

## 2. Generalized linear model in finite population

The literature concerning generalized linear models (GLM) is vast. [McCullagh and Nelder \(1989\)](#), [Dobson \(2001\)](#) and [Fox \(2008\)](#) are only a few examples of books presenting the subject. In a GLM setup,  $Y_1, \dots, Y_n$  are considered the values of an interest variable  $Y$  measured in  $n$  elements. The  $Y$ 's are supposed to be independent random variables with probability distribution belonging to the exponential family. Let  $Y_k$  be the random variable  $Y$  for element  $k$ . Its density function may be expressed as

$$f(y; \theta_k, \phi_k) = \exp\{\phi_k[y\theta_k - b(\theta_k)] + c(y, \phi_k)\}, \quad (1)$$

where  $c(\cdot)$  is a known function,  $\theta_k$  is the canonical parameter,  $E(Y_k) = \mu_k = b'(\theta_k)$  is the expected value,  $\text{Var}(Y_k) = \phi_k^{-1}V(\mu_k)$ , with  $V_k = V(\mu_k) = \partial \mu_k / \partial \theta_k$  is the variance function, and  $\phi_k^{-1} > 0$  is the dispersion parameter. Generalized linear models are defined by (1) and by the following systematic component

$$g(\mu_k) = \eta_k = \sum_{j=1}^J \beta_j x_{kj} = \mathbf{x}_k^T \boldsymbol{\beta}, \quad (2)$$

where  $\mathbf{x}_k = (x_{k1}, \dots, x_{kj})^T$  is a vector of  $J$  explanatory variables measured at element  $k$ ,  $\boldsymbol{\beta} = (\beta_1, \dots, \beta_J)^T$  a vector of unknown parameters, and  $g(\cdot)$ , a monotone differentiable function, called the link function. When  $g(\cdot)$  is defined in such a way that  $\theta_k = \eta_k$  for every  $k$ , then  $g(\cdot)$  is called the canonic link function. It has been assumed in this work that if  $k$  and  $l$  are such that  $\phi_k \neq \phi_l$ , then  $\phi_k \propto \delta_k$ , for  $k = 1, \dots, n$ , with  $\delta_k$  being a known quantity for every  $k$ . Particular cases of such setup include normal, Bernoulli, Poisson, gamma and the inverse Gaussian response models. Table 1 shows  $b(\theta)$ ,  $\theta$ ,  $\phi$  and  $V(\mu)$  forms for main exponential family distributions.

The maximum-likelihood method may be used to estimate GLM parameters. The estimation method can use sampling weights in the finite population context (see [Nordberg, 1989](#)). For instance,  $\hat{\mu}_k^U$  and  $\hat{\mu}_k^S$  are estimators of  $\mu_k$ ; however, the

**Table 2**  
Estimation of  $\beta$  and  $\mu_k$ .

|         | With information about $U$                             | With information about $S$                                             |                                                              |
|---------|--------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------|
| $\beta$ | $\hat{\beta}_U = \arg \max_{\beta} L_U(\beta)$         | Weighted<br>$\hat{\beta}_S^{\pi} = \arg \max_{\beta} L_S^{\pi}(\beta)$ | Unweighted<br>$\hat{\beta}_S = \arg \max_{\beta} L_S(\beta)$ |
| $\mu_k$ | $\hat{\mu}_k^U = g^{-1}(\mathbf{x}_k^T \hat{\beta}_U)$ | $\hat{\mu}_k^S = g^{-1}(\mathbf{x}_k^T \hat{\beta}_S^{\pi})$           | $\hat{\mu}_k^S = g^{-1}(\mathbf{x}_k^T \hat{\beta}_S)$       |

first one uses information about  $U$ , while the second one uses only information about  $S$ . Similarly, there are differences between  $\beta$ ,  $\hat{\beta}_U$ ,  $\hat{\beta}_S$  and  $\hat{\beta}_S^{\pi}$ , since  $\beta$  is the super-population parameter vector while  $\hat{\beta}_U$  is the  $\beta$  estimator based on  $U$ . There are two options for estimating  $\beta$  with information from sample  $S$ : the first, from which  $\hat{\beta}_S$  is obtained by assigning equal weighting to individuals; the second one, from which the pseudo-maximum likelihood estimator  $\hat{\beta}_S^{\pi}$  is obtained, takes into account sample design assigning weights for individuals according to first-order inclusion probabilities, (see Table 2).

The log-likelihood function for sample  $S$  ( $S \subset U$ ), considering sample weighting, often called  $\pi$ -weighted log-likelihood or pseudo-log-likelihood, can be written in the following way

$$L_S^{\pi}(\beta) = \sum_{k \in S} \frac{\phi_k}{\pi_k} [y_k \theta(\beta; \mathbf{x}_k) - b(\theta(\beta; \mathbf{x}_k))], \quad (3)$$

then,  $\hat{\beta}_S^{\pi} = \arg \max_{\beta} L_S^{\pi}(\beta)$  and  $\hat{\mu}_k^S = g^{-1}(\mathbf{x}_k^T \hat{\beta}_S^{\pi})$  are the pseudo-maximum likelihood estimators (see, for instance, Skinner and Holt, 1989; Lehtonen and Pahkinen, 2004) of  $\beta$  and  $\mu_k$ , respectively. If  $\pi_k = \pi$  for  $k \in U$ , then  $\hat{\beta}_S$  and  $\hat{\beta}_S^{\pi}$  are equivalents. The estimator  $\hat{\beta}_S^{\pi}$  can also be found by solving equation  $\mathbf{U}_S^{\pi}(\hat{\beta}_S^{\pi}) = \mathbf{0}$ . The function  $\mathbf{U}_S^{\pi}(\beta) = \partial L_S^{\pi}(\beta) / \partial \beta = (U_{S1}^{\pi}(\beta), \dots, U_{Sj}^{\pi}(\beta))^T$  is the score function for  $\beta$  (see, for example, Cox and Hinkley (1974)), where

$$U_{Sj}^{\pi}(\beta) = \sum_{k \in S} \frac{\phi_k}{\pi_k} (y_k - \mu_k) \frac{\delta \theta_k}{\delta \eta_k} x_{kj}, \quad j = 1, \dots, J. \quad (4)$$

The pseudo-maximum likelihood estimator is equivalent to the weighted least squares estimator for normal response models with canonical link (identity), which can be written in the following way

$$\hat{\beta}_S^{\pi} = (\mathbf{X}_S^T \mathbf{W}_S \mathbf{X}_S)^{-1} \mathbf{X}_S^T \mathbf{W}_S \mathbf{Y}_S, \quad (5)$$

where  $\mathbf{X}_S = (\mathbf{x}_1, \dots, \mathbf{x}_n)^T$ ,  $\mathbf{Y}_S = (y_1, \dots, y_n)^T$  and  $\mathbf{W}_S = \text{diag}\{\phi_1 \pi_1^{-1}, \dots, \phi_n \pi_n^{-1}\}$ . The log-likelihood function for the population  $U$  can be written as

$$L_U(\beta) = \sum_{k \in U} \phi_k [y_k \theta(\beta; \mathbf{x}_k) - b(\theta(\beta; \mathbf{x}_k))],$$

then,  $\hat{\beta}_U = \arg \max_{\beta} L_U(\beta)$  and  $\hat{\mu}_k^U = g^{-1}(\mathbf{x}_k^T \hat{\beta}_U)$  are the maximum likelihood estimators of  $\beta$  and  $\mu_k$ , respectively. The score function for  $\beta$  in the population  $U$  is given by  $\mathbf{U}_U(\beta) = \partial L_U(\beta) / \partial \beta = (U_{U1}(\beta), \dots, U_{Uj}(\beta))^T$ , where

$$U_{Uj}(\beta) = \sum_{k \in U} \phi_k (y_k - \mu_k) \frac{\delta \theta_k}{\delta \eta_k} x_{kj}, \quad j = 1, \dots, J. \quad (6)$$

### 3. Generalized linear model regression estimator (GEREG)

Consider the goal of estimating a population total,  $t_y = \sum_{k \in U} y_k$ . The GEREG for  $t_y$  may be written as

$$\hat{t}_G = \sum_{k \in U} \hat{\mu}_k^S + \sum_{k \in S} \frac{(y_k - \hat{\mu}_k^S)}{\pi_k}, \quad (7)$$

where  $\xi$ , the model explaining the relationship between interest and auxiliary variables, may be expressed as

$$\begin{cases} E_{\xi}(y_k) = \mu_k, \\ \text{Var}_{\xi}(y_k) = \phi_k^{-1} V(\mu_k), \\ g(\mu_k) = \sum_{j=1}^J \beta_j x_{kj} = \mathbf{x}_k^T \beta. \end{cases} \quad (8)$$

In the above model,  $\beta$  is a vector with unknown parameters,  $g(\cdot)$  is the link function,  $\mathbf{x}_k = (x_{k1}, \dots, x_{kj})^T$  is the auxiliary information vector for the  $k$ -th population element or cluster and  $E_{\xi}(\cdot)$  and  $\text{Var}_{\xi}(\cdot)$  are the expectation and variance under

model  $\xi$ , respectively. In expression (7)  $\hat{\mu}_k^S$  is given by  $g^{-1}(\mathbf{x}_k^T \hat{\boldsymbol{\beta}}_S^\pi)$ , with  $\hat{\boldsymbol{\beta}}_S^\pi$  as the  $\boldsymbol{\beta}$  estimator, based on  $S$ , taking sampling weighting into account. We assume that, if  $k$  and  $l$  are such that  $\phi_k \neq \phi_l$ , then  $\phi_k \propto \delta_k$ , with  $\delta_k$  being a known quantity for every  $k \in U$ . If  $\xi$  considers homogeneity in the dispersion parameter ( $\phi_k = \phi$ ), then we assume that  $\phi_k = 1$  for every  $k \in U$ . The  $\hat{t}_G$  estimator requires knowledge about auxiliary information vector  $\mathbf{x}_k$  for every  $k \in U$ . However, if the link function in model  $\xi$  is the identity,  $\hat{t}_G$  only requires knowledge about  $\mathbf{x}_k$  for every  $k \in S$  and population total vector  $\mathbf{t}_x = (t_{x1}, \dots, t_{xJ})^T$ , where  $t_{xj} = \sum_{k \in U} x_{kj}$ ,  $j = 1, \dots, J$ .

### 3.1. Alternative expression for GREG

The second term in (7) is an adjustment term. Theorem 1 provides conditions for which it can be eliminated, thus simplifying the expression for the GREG estimator. This theorem also allows writing  $t_y$  using the fitted values of the model based on  $U$ .

**Theorem 1.** *If the GREG estimator  $\hat{t}_G$  is based on a  $\xi$  model where it has been verified that  $\sum_{j=1}^J a_j \mathbf{h}_j = \mathbf{1}$  for  $\mathbf{h}_j = (h_{1j}, \dots, h_{Nj})^T$  with  $h_{kj} = \phi_k \frac{\partial \theta_k}{\partial \eta_k} x_{kj}$  and  $\mathbf{a} = (a_1, \dots, a_J)^T$  a constant vector, then*

- (1)  $\sum_{k \in U} (y_k - \hat{\mu}_k^U) = 0$ , i.e., the total of  $y$  can be written as  $t_y = \sum_{k \in U} \hat{\mu}_k^U$ ,
- (2)  $\sum_{k \in S} (y_k - \hat{\mu}_k^S) / \pi_k = 0$ , i.e., the estimator  $\hat{t}_G$  can be written as  $\hat{t}_G = \sum_{k \in U} \hat{\mu}_k^S$ .

According to Theorem 1, the smaller the difference between  $\hat{\boldsymbol{\beta}}_S^\pi$  and  $\hat{\boldsymbol{\beta}}_U$  the smaller also the difference between  $\hat{t}_G$  and  $t_y$ . Also, since  $\hat{\boldsymbol{\beta}}_S^\pi$  is a consistent estimator for  $\hat{\boldsymbol{\beta}}_U$ ,  $\hat{t}_G$  is consistent for  $t_y$ . Moreover, if  $\pi_k = \pi$  for  $k \in U$ , and the conditions of Theorem 1 are satisfied, then we can write  $\hat{t}_G = \sum_{k \in S} y_k + \sum_{k \in S^*} \hat{\mu}_k^S$ , where  $S^*$  represents the set of unobserved units or clusters in the finite population.

**Corollary 1.** *If GREG estimator  $\hat{t}_G$  is based on a  $\xi$  model where it has been verified that there is*

- (1) Homogeneity in the dispersion parameter, i.e.,  $\phi_k = \phi$  for  $k \in U$ ;
- (2) A systematic component having an intercept, i.e., there is  $\beta_j$  in  $\boldsymbol{\beta}$  so that  $\partial \eta_k / \partial \beta_j = c$  for  $k \in U$ , with  $c$  a constant;
- (3) A systematic component having a canonical link, i.e.,  $\theta_k = \eta_k$  for  $k \in U$ ;

then, the results (1) and (2) of Theorem 1 are satisfied.

The Corollary 1 provides, from the model formulation point of view, clearer conditions than the Theorem 1 so that the adjustment term in the GREG estimator can be eliminated.

**Corollary 2.** *If the GREG estimator  $\hat{t}_G$  is based on an  $\xi$  model with normal response, canonical link (identity) and  $\mathbf{a} = (a_1, \dots, a_J)^T$  is a constant vector so that, for  $k \in U$  it has been verified that*

$$\phi_k^{-1} = \sum_{j=1}^J a_j x_{kj},$$

then, the results (1) and (2) of Theorem 1 are satisfied.

Corollary 2 results is analogous to that found in Särndal et al. (2003, p. 230). Hence, Theorem 1 may be viewed as an extension of the result shown in Särndal et al. (2003) for generalized linear models.

### 3.2. Expectation and variance

The following theorem provides approximate expressions for the expectation and variance of the GREG estimator, taking into account the sampling design and the formulated regression model.

**Theorem 2.** *If  $\hat{t}_G$  is given by  $\sum_{k \in U} \hat{\mu}_k^S$  and  $\xi$  considers the canonical link then*

- (1)  $E(\hat{t}_G) \approx t_y$ ;
- (2)  $\text{Var}(\hat{t}_G) \approx \Gamma_U^T \Sigma_U \Gamma_U$ ;
- (3)  $\hat{\text{Var}}(\hat{t}_G) = \Gamma_S^T \Sigma_S \Gamma_S$ ,

where  $\Gamma_U = (\mathbf{X}_U^T \mathbf{W}_U^* \mathbf{X}_U)^{-1} \mathbf{X}_U^T \mathbf{W}_U^* \boldsymbol{\phi}^*$ ,  $\mathbf{W}_U^* = \text{diag}\{\phi_1 \hat{V}_1^U, \dots, \phi_N \hat{V}_N^U\}$ ,  $\Gamma_S = (\mathbf{X}_U^T \mathbf{W}_S^* \mathbf{X}_U)^{-1} \mathbf{X}_U^T \mathbf{W}_S^* \boldsymbol{\phi}^*$ ,  $\mathbf{W}_S^* = \text{diag}\{\phi_1 \hat{V}_1^S, \dots, \phi_N \hat{V}_N^S\}$ ,  $\Sigma_U = \{\sigma_{ij}^U\}$  and  $\Sigma_S = \{\sigma_{ij}^S\}$ , with  $\sigma_{ij}^U = \sum_{k \in U} \sum_{l \in U} \Delta_{kl} \frac{x_{ki} \phi_k E_k}{\pi_k} \frac{x_{lj} \phi_l E_l}{\pi_l}$ ,  $\sigma_{ij}^S = \sum_{k \in S} \sum_{l \in S} \frac{\Delta_{kl}}{\pi_{kl}} \frac{x_{ki} \phi_k e_k}{\pi_k} \frac{x_{lj} \phi_l e_l}{\pi_l}$ ,  $\boldsymbol{\phi}^* = (\phi_1^{-1}, \dots, \phi_N^{-1})^T$ ,  $\hat{V}_U = V(\hat{\mu}_k^U)$ ,  $\hat{V}_S = V(\hat{\mu}_k^S)$ ,  $\Delta_{kl} = \pi_{kl} - \pi_k \pi_l$ ,  $E_k = y_k - \hat{\mu}_k^U$  and  $e_k = y_k - \hat{\mu}_k^S$ .

**Corollary 3.** Under the conditions of the [Corollary 1](#), the following results hold:

- (1)  $E(\hat{t}_G) \approx t_y$ ;
- (2)  $\text{Var}(\hat{t}_G) \approx \sum_{k \in U} \sum_{l \in U} \Delta_{kl} \frac{E_k}{\pi_k} \frac{E_l}{\pi_l}$ ;
- (3)  $\hat{\text{Var}}(\hat{t}_G) = \sum_{k \in S} \sum_{l \in S} \frac{\Delta_{kl}}{\pi_{kl}} \frac{e_k}{\pi_k} \frac{e_l}{\pi_l}$ .

**Corollary 4.** If the GREG estimator  $\hat{t}_G$  is based on a  $\xi$  model with normal response, canonical link (identity) and  $\mathbf{a} = (a_1, \dots, a_j)^T$  is a constant vector so that, for  $k \in U$  it has been verified that

$$\phi_k^{-1} = \sum_{j=1}^J a_j x_{kj},$$

then, the results (1)–(3) of [Corollary 3](#) are satisfied.

[Corollary 3](#) shows that when the  $\xi$  model considers canonical link, intercept and homogeneity in the dispersion parameter, the expressions for  $\text{Var}(\hat{t}_G)$  and  $\hat{\text{Var}}(\hat{t}_G)$  become simplified, i.e., stated that the estimator  $\hat{t}_G$  is approximately unbiased for  $t_y$  and, an approximation to the variance of  $\hat{t}_G$  can be calculated applying the formula for the variance of the Horvitz–Thompson estimator to the residuals  $E_k$ .

### 3.3. The model's role

The role of model  $\xi$  in estimating  $t_y$  can be observed by assessing the impact of small perturbations in  $y_k$ , on  $E_k$ . [Pregibon \(1981\)](#) proposed a quantity for measuring the influence of  $y_k$  on  $\hat{\mu}_k^U$  in generalized linear models, this being an extension of the measurement  $\partial \hat{\mu}_k^U / \partial y_k$  applied to normal linear models. Then, when  $\xi$  considers canonical link and homogeneity in dispersion parameter, the quantity  $\gamma_k = 1 - \hat{h}_{kk}$  can be used to assess the influence of  $y_k$  on  $E_k$ , where  $\hat{h}_{kk}$  is the main diagonal  $k$ -th element of  $\hat{\mathbf{H}} = \hat{\mathbf{V}}_U^{1/2} \mathbf{X}_U (\mathbf{X}_U^T \hat{\mathbf{V}}_U \mathbf{X}_U)^{-1} \mathbf{X}_U^T \hat{\mathbf{V}}_U^{1/2}$ , with  $\hat{\mathbf{V}}_U = \text{diag}\{\hat{V}_1^U, \dots, \hat{V}_N^U\}$ . Small  $\gamma_k$  values indicate that a change in  $y_k$  leads to a similar change on  $\hat{\mu}_k^U$ , suggesting that the  $k$ -th population element has a large weight on estimating  $\hat{\beta}^U$  and on the  $E_k$  magnitude. Likewise, large  $\gamma_k$  values indicate that a change in  $y_k$  does not similarly lead to a change in  $\hat{\mu}_k^U$ , suggesting that the  $k$ -th population element has small weight in estimating  $\hat{\beta}^U$  and on the  $E_k$  magnitude. For example, for a model having an intercept, one explanatory variable and homogeneity in the dispersion parameter, we have the following

(i) Canonical link

$$\gamma_k = 1 - \frac{\hat{V}_k^U}{\sum_{l \in U} \hat{V}_l^U} - \frac{(x_k - \bar{x}_U)^2 \hat{V}_k^U}{\sum_{l \in U} (x_l - \bar{x}_U)^2 \hat{V}_l^U}.$$

(ii) Identity link

$$\gamma_k = 1 - \frac{(1/\hat{V}_k^U)}{\sum_{l \in U} (1/\hat{V}_l^U)} - \frac{(x_k - \bar{x}_U)^2 (1/\hat{V}_k^U)}{\sum_{l \in U} (x_l - \bar{x}_U)^2 (1/\hat{V}_l^U)},$$

where  $\bar{x}_U$  is the population mean of the explanatory variable  $x$ . In both cases considered above,  $\gamma_k \in (0, 1)$  and, for a fixed value of  $x_k$ , the  $\gamma_k$  value can be directly proportional to the variance function, when the link is canonical, or it can be inversely proportional to the variance function, when the link is the identity. Thus, the  $\xi$  model determines the structure of weighting or importance for elements in  $U$ , which affects the values of  $E_k$  on  $\text{Var}(\hat{t}_G)$ .

### 3.4. Relative efficiency

The relative efficiency of  $\hat{t}_G$  compared to the Horvitz–Thompson estimator using the  $p(\cdot)$  sampling design can be calculated using the measure  $\Delta_{p(\cdot)}$ , which has been defined as  $\Delta_{p(\cdot)} = \text{Var}(\hat{t}_G)_{p(\cdot)} / \text{Var}(\hat{t}_{HT})_{p(\cdot)}$ . For example, for sampling designs such as simple random sampling (SRS) and Bernoulli sampling (BER) (see, for instance, [Särndal et al., 2003](#), ch. 3) the  $\Delta_{p(\cdot)}$  measurement does not depend on the sample size. The following result led us to observe the impact of the goodness of fit of  $\xi$  on the relative efficiency of  $\hat{t}_G$ .

**Theorem 3.** Under the conditions of the [Corollary 1](#), if the  $\hat{t}_G$  is used under

(i) Simple Random Sampling (SRS), then

$$\Delta_{\text{SRS}} = \frac{\text{Var}(\hat{t}_G)_{\text{SRS}}}{\text{Var}(\hat{t}_{HT})_{\text{SRS}}} \approx 1 - R_U^2.$$

(ii) *Bernoulli Sampling (BER)*, then

$$\Delta_{BER} = \frac{\text{Var}(\hat{t}_G)_{BER}}{\text{Var}(\hat{t}_{HT})_{BER}} \approx \frac{1 - R_U^2}{1 + \frac{N}{(N-1)CV_y^2}},$$

where  $R_U$  is the Pearson's linear correlation coefficient calculated between  $y_k$ 's and  $\hat{\mu}_k^U$ 's, and  $CV_y$  is the  $y$  coefficient of variation in the population  $U$ .

This result shows that when the model's goodness of fit increases then the  $\hat{t}_G$  efficiency compared to the Horvitz–Thompson estimator also increases. It is also possible to observe that, using comparable sample sizes, we have  $\text{Var}(\hat{t}_G)_{BER}/\text{Var}(\hat{t}_G)_{SRS} \approx 1 - 1/N$ . Further, [Theorem 3](#) means that it is possible to show that under SRS sampling, the Horvitz–Thompson estimator has efficiency equal to  $\hat{t}_G$  when using a sampling fraction  $f^*$  given by

$$f^* \approx \frac{f}{f + (1-f)(1 - R_U^2)},$$

where  $f = n/N$  is the sampling fraction used by  $\hat{t}_G$ . For instance, when the GREG estimator uses a 0.2 sampling fraction under SRS sampling, the  $\xi$  model satisfies [Corollary 1](#) conditions and has a goodness of fit described by  $R_U^2 = 0.95$ , the Horvitz–Thompson estimator requires a sampling fraction around four times greater than the GREG to obtaining the same efficiency.

### 3.5. Particular cases

This section considers estimators assisted by models satisfying the [Theorem 1](#) conditions for different response distributions.

#### 3.5.1. Normal response with variance homogeneity

The estimator (9) is obtained through the least squares method in the statistical literature, with no distributional assumptions. The same estimator was obtained in this paper using a regression model with normal response and canonical link. According to [Corollary 1](#), this estimator is given by

$$\hat{t}_G = \sum_{k \in U} \hat{\mu}_k^S = \sum_{k \in U} \mathbf{x}_k^T \hat{\boldsymbol{\beta}}_S^\pi = N \hat{\beta}_{S1}^\pi + \sum_{j=2}^J \hat{\beta}_{Sj}^\pi t_{x_j}, \quad (9)$$

with  $\hat{\boldsymbol{\beta}}_S^\pi = (\hat{\beta}_{S1}^\pi, \dots, \hat{\beta}_{Sj}^\pi)^T$ . The model  $\xi$  that assists (9) can be written as

$$\begin{cases} E_\xi(Y_k) = \mu_k, \\ \text{Var}_\xi(Y_k) = \sigma^2, \\ \mu_k = \beta_1 + \sum_{j=2}^J \beta_j x_{kj}. \end{cases}$$

From (5)  $\hat{\boldsymbol{\beta}}_S^\pi$  can be written as  $\hat{\boldsymbol{\beta}}_S^\pi = (\mathbf{X}_S^T \mathbf{W}_S \mathbf{X}_S)^{-1} \mathbf{X}_S^T \mathbf{W}_S \mathbf{Y}_S$ , where  $\mathbf{x}_k = (1, x_{k2}, \dots, x_{kj})^T$  and  $\mathbf{W}_S = \text{diag}\{1/\pi_1, \dots, 1/\pi_n\}$ . According to [Corollary 3](#) under SRS sampling, the approximated variance of (9) and its corresponding estimator can be written as

$$\text{Var}(\hat{t}_G) \approx N(1 - R_U^2) \frac{1-f}{f} S_{yU}^2 \quad \text{and} \quad \hat{\text{Var}}(\hat{t}_G) = N(1 - R_S^2) \frac{1-f}{f} S_{yS}^2,$$

respectively, where  $S_{yU}^2 = \frac{1}{N-1} \sum_U (y_k - \bar{y}_U)^2$ ,  $S_{yS}^2 = \frac{1}{n-1} \sum_S (y_k - \bar{y}_S)^2$  and  $R_S$  is the Pearson's linear correlation coefficient calculated between  $y_k$ 's and  $\hat{\mu}_k^S$ 's. Similarly, from [Corollary 3](#), for BER sampling,

$$\text{Var}(\hat{t}_G) \approx \frac{N(1 - R_U^2)}{(1 - 1/N)^{-1}} \frac{1 - \pi}{\pi} S_{yU}^2 \quad \text{and} \quad \hat{\text{Var}}(\hat{t}_G) = \frac{n}{N\pi} \frac{N(1 - R_S^2)}{(1 - 1/n)^{-1}} \frac{1 - \pi}{\pi} S_{yS}^2,$$

where  $N\pi$  and  $n$  are expected and effective sample sizes, respectively.

#### 3.5.2. Normal response with variance heterogeneity

The following estimator satisfies the conditions of [Corollary 2](#), and was assisted by a model with normal response and variance heterogeneity

$$\hat{t}_G = \sum_{k \in U} \hat{\mu}_k^S = \sum_{k \in U} \hat{\beta}_S^\pi x_k = \hat{\beta}_S^\pi t_x = \frac{t_x}{\hat{t}_x} \hat{t}_y, \quad (10)$$

with  $\hat{t}_x$  and  $\hat{t}_y$  being the Horvitz–Thompson estimators for  $t_x$  and  $t_y$ , respectively. The estimator (10) is the well known *ratio estimator* (see for instance, [Särndal et al., 2003](#), p. 247). The  $\xi$  model that assisted (10) is given by

$$\begin{cases} E_{\xi}(Y_k) = \mu_k, \\ \text{Var}_{\xi}(Y_k) = \sigma^2 x_k, & x_k > 0, \\ \mu_k = \beta x_k. \end{cases}$$

From (5)  $\hat{\beta}_S^{\pi}$  can be written as  $\hat{\beta}_S^{\pi} = (\mathbf{X}_S^T \mathbf{W}_S \mathbf{X}_S)^{-1} \mathbf{X}_S^T \mathbf{W}_S \mathbf{Y}_S$  with  $\mathbf{W}_S = \text{diag}\{1/x_1\pi_1, \dots, 1/x_n\pi_n\}$ . According to [Theorem 2](#) under SRS sampling, the approximated variance of (10) and its corresponding estimator are given by

$$\text{Var}(\hat{t}_G) = N \frac{1-f}{f} \left[ S_{y_U}^2 + \frac{S_{x_U} S_{y_U} \bar{y}_U^2}{\bar{x}_U^2} \left( \frac{S_{x_U}}{S_{y_U}} - 2R_U \right) \right]$$

and

$$\hat{\text{Var}}(\hat{t}_G) = N \frac{1-f}{f} \left[ S_{y_s}^2 + \frac{S_{x_s} S_{y_s} \bar{y}_s^2}{\bar{x}_s^2} \left( \frac{S_{x_s}}{S_{y_s}} - 2R_s \right) \right],$$

respectively. Similarly, from [Theorem 2](#), under BER sampling,

$$\text{Var}(\hat{t}_G) = \frac{N}{(1-1/N)^{-1}} \frac{1-\pi}{\pi} \left[ S_{y_U}^2 + \frac{S_{x_U} S_{y_U} \bar{y}_U^2}{\bar{x}_U^2} \left( \frac{S_{x_U}}{S_{y_U}} - 2R_U \right) \right]$$

and

$$\hat{\text{Var}}(\hat{t}_G) = \frac{n}{N\pi} \frac{N}{(1-1/n)^{-1}} \frac{1-\pi}{\pi} \left[ S_{y_s}^2 + \frac{S_{x_s} S_{y_s} \bar{y}_s^2}{\bar{x}_s^2} \left( \frac{S_{x_s}}{S_{y_s}} - 2R_s \right) \right].$$

One can easily show that, using SRS and BER sampling designs,  $\hat{t}_G$  is more efficient than the Horvitz–Thompson estimator when  $2R_U > S_{x_U}/S_{y_U}$ .

### 3.5.3. Bernoulli response

Suppose  $y_k$ , the response for element  $k$  in a particular population is a dichotomous variable, assuming value one if the element presents a characteristic of interest, and zero, otherwise. The  $t_y$  parameter of interest can be written as  $NP$ , where  $N$  is the population size and  $P$  is the proportion of population elements having the characteristic of interest. The estimator for  $NP$ , assisted by a Bernoulli regression model  $\xi$  satisfying the conditions of [Corollary 1](#) can be written as

$$\hat{t}_G = \sum_{k \in U} \hat{\mu}_k^S = \sum_{k \in U} \left[ 1 + \exp \left( -\hat{\beta}_{S1}^{\pi} - \sum_{j=2}^J \hat{\beta}_{Sj}^{\pi} x_{kj} \right) \right]^{-1}, \quad (11)$$

where  $\xi$ , the model assisting in the estimation, can be described as

$$\begin{cases} E_{\xi}(Y_k) = \mu_k = P_{\xi}[y_k = 1 | \mathbf{x}_k], \\ \text{Var}_{\xi}(Y_k) = \mu_k(1 - \mu_k), \\ \log \left( \frac{\mu_k}{1 - \mu_k} \right) = \beta_1 + \sum_{j=2}^J \beta_j x_{kj}. \end{cases}$$

As discussed in [Section 3.3](#), for a given  $\mathbf{x}_k$ , the closer  $\hat{\mu}_k$  is of 0.5, the greater the weight or importance of element  $k$  in the estimation. According to [Corollary 3](#), under SRS sampling, the approximated variance of (11) and its corresponding estimator can be expressed as

$$\text{Var}(\hat{t}_G) \approx \frac{N(1-R_U^2)}{1-1/N} \frac{1-f}{f} P(1-P) \quad \text{and} \quad \hat{\text{Var}}(\hat{t}_G) = \frac{N(1-R_s^2)}{1-1/n} \frac{1-f}{f} \hat{P}(1-\hat{P}),$$

where  $\hat{P} = \frac{1}{n} \sum_s y_k$  is the sample proportion of elements having the characteristic of interest. On the other hand, in BER sampling, the approximated variance of (11) and its corresponding estimator can be expressed as

$$\text{Var}(\hat{t}_G) \approx N(1-R_U^2) \frac{1-\pi}{\pi} P(1-P) \quad \text{and} \quad \hat{\text{Var}}(\hat{t}_G) = \frac{n}{N\pi} N(1-R_s^2) \frac{1-\pi}{\pi} \hat{P}(1-\hat{P}).$$



### 3.5.4. Poisson response

When the variable of interest represents a counting process, estimating population totals can be assisted by a Poisson regression model. Such an estimator, which satisfies the conditions of [Corollary 1](#), can be expressed by

$$\hat{t}_G = \sum_{k \in U} \hat{\mu}_k^S = \sum_{k \in U} \exp \left( \hat{\beta}_{S1}^\pi + \sum_{j=2}^J \hat{\beta}_{Sj}^\pi x_{kj} \right) = \hat{\kappa}_1^\pi \sum_{k \in U} \prod_{j=2}^J \hat{\kappa}_j^{\pi x_{kj}}, \quad (12)$$

with  $\hat{\kappa}_j^\pi = \exp(\hat{\beta}_{Sj}^\pi)$ , where  $\xi$ , the model assisting the estimation, is given by

$$\begin{cases} E_\xi(Y_k) = \mu_k, \\ \text{Var}_\xi(Y_k) = \mu_k, \\ \log(\mu_k) = \beta_1 + \sum_{j=2}^J \beta_j x_{kj}. \end{cases}$$

As discussed in Section 3.3, for a given  $\mathbf{x}_k$ , the higher  $\hat{\mu}_k$ , the higher the relevance (weight) of element  $k$ . According to [Corollary 3](#), in SRS sampling, the approximated variance of (12) and its corresponding estimator can be expressed as

$$\text{Var}(\hat{t}_G) \approx N(1 - R_U^2) \frac{1-f}{f} S_{yU}^2 \quad \text{and} \quad \hat{\text{Var}}(\hat{t}_G) = N(1 - R_S^2) \frac{1-f}{f} S_{yS}^2.$$

On the other hand, in BER sampling, the approximated variance of (12) and its corresponding estimator can be expressed as

$$\text{Var}(\hat{t}_G) \approx \frac{N(1 - R_U^2)}{(1 - 1/N)^{-1}} \frac{1-\pi}{\pi} S_{yU}^2 \quad \text{and} \quad \hat{\text{Var}}(\hat{t}_G) = \frac{n}{N\pi} \frac{N(1 - R_S^2)}{(1 - 1/n)^{-1}} \frac{1-\pi}{\pi} S_{yS}^2.$$

### 3.5.5. Gamma response

When the variable of interest is skewed continuous and positive (i.e., income) the estimation can be assisted by the gamma model. The estimator of  $t_y$  assisted by a gamma response regression model, and satisfying the conditions of [Corollary 1](#) can be expressed in the following way

$$\hat{t}_G = \sum_{k \in U} \hat{\mu}_k^S = \sum_{k \in U} \left( \hat{\beta}_{S1}^\pi + \sum_{j=2}^J \hat{\beta}_{Sj}^\pi x_{kj} \right)^{-1}, \quad (13)$$

where  $\xi$ , the model that assisting the estimation, can be written as

$$\begin{cases} E_\xi(Y_k) = \mu_k, \\ \text{Var}_\xi(Y_k) = \mu_k^2, \\ \mu_k^{-1} = \beta_1 + \sum_{j=2}^J \beta_j x_{kj}. \end{cases}$$

As discussed in Section 3.3, for a given  $\mathbf{x}_k$ , the higher  $\hat{\mu}_k$ , the higher the relevance (weight) of element  $k$ . According to [Corollary 3](#), in SRS sampling, the approximated variance of (13) and its corresponding estimator are given by

$$\text{Var}(\hat{t}_G) \approx N(1 - R_U^2) \frac{1-f}{f} S_{yU}^2 \quad \text{and} \quad \hat{\text{Var}}(\hat{t}_G) = N(1 - R_S^2) \frac{1-f}{f} S_{yS}^2.$$

On the other hand, in BER sampling, the approximated variance of (13) and its corresponding estimator can be expressed as

$$\text{Var}(\hat{t}_G) \approx \frac{N(1 - R_U^2)}{(1 - 1/N)^{-1}} \frac{1-\pi}{\pi} S_{yU}^2 \quad \text{and} \quad \hat{\text{Var}}(\hat{t}_G) = \frac{n}{N\pi} \frac{N(1 - R_S^2)}{(1 - 1/n)^{-1}} \frac{1-\pi}{\pi} S_{yS}^2.$$

### 3.5.6. Stratified sampling

(i) Estimating population totals.

For a stratified SRS design in which the model assisting estimation in each stratum satisfies the conditions of [Corollary 1](#), the estimator for  $t_y$  is

$$\hat{t}_G = \sum_{h=1}^H \hat{t}_{Gh} = \sum_{h=1}^H \sum_{k \in U_h} \hat{\mu}_k^{S_h},$$

where  $\hat{t}_{Gh}$  is the GREG estimator of the total of  $y$  in stratum  $h$ . According to [Corollary 3](#), the variance estimator expression is given by

$$\text{Var}(\hat{t}_G) = \sum_{h=1}^H \text{Var}(\hat{t}_{Gh}) \approx \sum_{h=1}^H N_h(1 - R_{U_h}^2) \frac{1-f_h}{f_h} S_{yU_h}^2,$$



with  $N_h$  being the size of the stratum  $h$ ,  $f_h$  the sampling fraction in the stratum  $h$ ,  $S_{y_{U_h}}^2$  the variance of  $y$  in the stratum  $h$  and  $R_{U_h}$  is Pearson's linear correlation coefficient calculated between  $y_k$ 's and  $\hat{\mu}_k^{U_h}$ 's in the stratum  $h$ . The optimum sample allocation under a linear cost function is achieved by choosing  $n_h$  in the following way

$$n_h = n \frac{N_h \sqrt{1 - R_{U_h}^2 S_{y_{U_h}}^2 / c_h}}{\sum_{i=1}^H N_i \sqrt{1 - R_{U_i}^2 S_{y_{U_i}}^2 / c_i}},$$

where  $c_h > 0$  is the cost to sample an element or cluster from the stratum  $h$ . The estimator of the variance is given by

$$\hat{\text{Var}}(\hat{t}_G) = \sum_{h=1}^H \hat{\text{Var}}(\hat{t}_{G_h}) = \sum_{h=1}^H N_h (1 - R_{S_h}^2) \frac{1 - f_h}{f_h} S_{y_{S_h}}^2,$$

where  $S_{y_{S_h}}^2$  is the variance of  $y$  in the sample of stratum  $h$  and  $R_{S_h}$  is Pearson's linear correlation coefficient calculated between  $y_k$ 's and  $\hat{\mu}_k^{S_h}$ 's in the sample from stratum  $h$ . With BER sampling, the total population estimator expression is the same as for the SRS case. The variance estimator expression can be written as

$$\text{Var}(\hat{t}_G) = \sum_{h=1}^H \text{Var}(\hat{t}_{G_h}) \approx \sum_{h=1}^H (N_h - 1) (1 - R_{U_h}^2) \frac{1 - \pi_h}{\pi_h} S_{y_{U_h}}^2,$$

with  $N_h \pi_h$  being expected sample size in stratum  $h$ . The variance estimator expression is given by

$$\hat{\text{Var}}(\hat{t}_G) = \sum_{h=1}^H \hat{\text{Var}}(\hat{t}_{G_h}) = \sum_{h=1}^H \frac{(n_h - 1)}{\pi_h} (1 - R_{S_h}^2) \frac{1 - \pi_h}{\pi_h} S_{y_{S_h}}^2,$$

where  $n_h$  is the sample size in stratum  $h$ .

(ii) Estimating population means.

For a stratified SRS design in which the model assisting estimation in each stratum satisfies the conditions of [Corollary 1](#), the population mean estimator is given by

$$\hat{m}_G = \sum_{h=1}^H a_h \hat{m}_{G_h} = \sum_{h=1}^H a_h \sum_{k \in U_h} N_h^{-1} \hat{\mu}_k^{S_h},$$

with  $\hat{m}_{G_h}$  being the GREG estimator of mean in stratum  $h$  and  $a_h = N_h / (\sum N_h)$ , the proportion of individuals from population in stratum  $h$ . According to [Corollary 3](#), the variance estimator expression is given by

$$\text{Var}(\hat{m}_G) = \sum_{h=1}^H a_h^2 \text{Var}(\hat{m}_{G_h}) \approx \sum_{h=1}^H a_h^2 \frac{1 - R_{U_h}^2}{N_h} \frac{1 - f_h}{f_h} S_{y_{U_h}}^2.$$

The optimum sample allocation under a linear cost function is achieved by choosing  $n_h$  in the following way

$$n_h = n \frac{a_h \sqrt{1 - R_{U_h}^2 S_{y_{U_h}}^2 / c_h}}{\sum_{i=1}^H a_i \sqrt{1 - R_{U_i}^2 S_{y_{U_i}}^2 / c_i}}.$$

The variance estimator expression is given by

$$\hat{\text{Var}}(\hat{m}_G) = \sum_{h=1}^H a_h^2 \hat{\text{Var}}(\hat{m}_{G_h}) = \sum_{h=1}^H a_h^2 \frac{1 - \hat{R}_{S_h}^2}{N_h} \frac{1 - f_h}{f_h} S_{y_{S_h}}^2.$$

With BER sampling, the mean population estimator expression is the same as for the SRS case. The variance expression of the estimator is given by

$$\text{Var}(\hat{m}_G) = \sum_{h=1}^H a_h^2 \text{Var}(\hat{m}_{G_h}) \approx \sum_{h=1}^H a_h^2 \frac{N_h - 1}{N_h^2} (1 - R_{U_h}^2) \frac{1 - \pi_h}{\pi_h} S_{y_{U_h}}^2.$$

The variance is estimated by

$$\hat{\text{Var}}(\hat{m}_G) = \sum_{h=1}^H a_h^2 \hat{\text{Var}}(\hat{m}_{G_h}) = \sum_{h=1}^H a_h^2 \frac{(n_h - 1)}{N_h^2 \pi_h} (1 - R_{S_h}^2) \frac{1 - \pi_h}{\pi_h} S_{y_{S_h}}^2.$$

### 3.6. Simulation study

A simulation study was carried out for comparing the behavior of the estimators described above in different scenarios.

**Table 3**Performance of  $\hat{P}_N$  and  $\hat{P}_B$  estimating the population proportion  $P$ .

| $R_U^2$ | $n$  | $P$  | Measures (%)                           |                                      |                                                          |       |                                        |                                      |                                                          |       |
|---------|------|------|----------------------------------------|--------------------------------------|----------------------------------------------------------|-------|----------------------------------------|--------------------------------------|----------------------------------------------------------|-------|
|         |      |      | $\hat{P}_N$                            |                                      |                                                          |       | $\hat{P}_B$                            |                                      |                                                          |       |
|         |      |      | $\left  \frac{\hat{P}}{P} - 1 \right $ | $\frac{V(\hat{P})}{V(\hat{P}_{HT})}$ | $\left  \frac{\hat{V}(\hat{P})}{V(\hat{P})} - 1 \right $ | CR    | $\left  \frac{\hat{P}}{P} - 1 \right $ | $\frac{V(\hat{P})}{V(\hat{P}_{HT})}$ | $\left  \frac{\hat{V}(\hat{P})}{V(\hat{P})} - 1 \right $ | CR    |
| 0.50    | 40   | 0.15 | 1.51                                   | 65.18                                | 8.39                                                     | 91.81 | 0.86                                   | 54.11                                | 13.35                                                    | 89.06 |
|         |      | 0.35 | 0.09                                   | 58.29                                | 5.99                                                     | 93.55 | 0.12                                   | 52.32                                | 6.35                                                     | 92.29 |
|         |      | 0.65 | 0.74                                   | 65.93                                | 11.23                                                    | 92.52 | 0.13                                   | 53.75                                | 12.76                                                    | 91.18 |
|         |      | 0.85 | 0.30                                   | 76.18                                | 3.51                                                     | 92.01 | 0.30                                   | 54.60                                | 11.57                                                    | 90.36 |
|         | 50   | 0.15 | 1.03                                   | 65.57                                | 8.12                                                     | 92.16 | 0.77                                   | 54.29                                | 12.85                                                    | 90.19 |
|         |      | 0.35 | 0.10                                   | 60.04                                | 7.95                                                     | 93.30 | 0.08                                   | 53.75                                | 7.55                                                     | 92.58 |
|         |      | 0.65 | 0.72                                   | 63.14                                | 6.58                                                     | 93.62 | 0.08                                   | 51.97                                | 8.98                                                     | 92.28 |
|         |      | 0.85 | 0.30                                   | 74.83                                | 1.52                                                     | 92.52 | 0.19                                   | 52.81                                | 9.43                                                     | 90.60 |
|         | 0.65 | 0.15 | 1.83                                   | 59.21                                | 10.73                                                    | 91.49 | 0.84                                   | 41.79                                | 20.19                                                    | 85.12 |
|         |      | 0.35 | 0.13                                   | 48.17                                | 8.20                                                     | 93.27 | 0.03                                   | 37.31                                | 11.81                                                    | 90.53 |
| 0.75    | 40   | 0.65 | 0.96                                   | 58.66                                | 10.85                                                    | 92.49 | 0.12                                   | 37.76                                | 12.19                                                    | 90.19 |
|         |      | 0.85 | 0.35                                   | 72.45                                | 3.65                                                     | 93.21 | 0.30                                   | 37.88                                | 6.72                                                     | 90.97 |
|         | 50   | 0.15 | 1.22                                   | 57.08                                | 6.61                                                     | 92.64 | 0.76                                   | 38.88                                | 14.32                                                    | 89.01 |
|         |      | 0.35 | 0.25                                   | 48.37                                | 8.05                                                     | 93.36 | 0.11                                   | 37.23                                | 10.87                                                    | 91.13 |
|         |      | 0.65 | 0.75                                   | 57.93                                | 9.03                                                     | 93.13 | 0.01                                   | 37.69                                | 11.68                                                    | 91.14 |
|         |      | 0.85 | 0.24                                   | 69.75                                | 0.39                                                     | 93.46 | 0.20                                   | 36.78                                | 6.91                                                     | 90.68 |
|         | 0.85 | 0.15 | 2.70                                   | 57.40                                | 12.75                                                    | 90.43 | 0.04                                   | 32.76                                | 23.97                                                    | 80.00 |
|         |      | 0.35 | 0.45                                   | 42.59                                | 5.66                                                     | 93.52 | 0.23                                   | 27.58                                | 11.01                                                    | 90.11 |
|         |      | 0.65 | 0.99                                   | 55.48                                | 10.63                                                    | 92.79 | 0.07                                   | 32.62                                | 12.27                                                    | 90.25 |
|         |      | 0.85 | 0.38                                   | 72.24                                | 4.16                                                     | 92.63 | 0.29                                   | 32.86                                | 8.02                                                     | 90.87 |
| 0.85    | 40   | 0.15 | 1.42                                   | 54.30                                | 6.21                                                     | 92.82 | 0.42                                   | 29.38                                | 14.39                                                    | 86.96 |
|         |      | 0.35 | 0.22                                   | 43.57                                | 8.09                                                     | 93.34 | 0.02                                   | 28.00                                | 14.17                                                    | 89.79 |
|         |      | 0.65 | 0.80                                   | 54.38                                | 8.28                                                     | 93.10 | 0.05                                   | 31.68                                | 9.94                                                     | 91.38 |
|         |      | 0.85 | 0.22                                   | 68.89                                | 0.78                                                     | 93.34 | 0.17                                   | 30.34                                | 5.44                                                     | 90.83 |

### 3.6.1. Bernoulli response

Twelve populations consisting of  $N = 1000$  individuals each were generated where one auxiliary variable,  $x$ , was considered, generated by a chi-square distribution with 1 degree of freedom. The value of  $\hat{\mu}_k^U$  for the element  $k$  from each population was calculated as  $\hat{\mu}_k^U = \left[ 1 + \exp(-\hat{\beta}_{U_1} - \hat{\beta}_{U_2}x_k) \right]^{-1}$ , and the interest variable,  $y_k$ , was randomly generated following a Bernoulli distribution with parameter  $\hat{\mu}_k^U$ . In each population, the  $\hat{\beta}_U = (\hat{\beta}_{U_1}, \hat{\beta}_{U_2})^T$  values were chosen so that  $R_U^2 = 0.5, 0.65$  and  $0.75$ , and  $P = 0.15, 0.35, 0.65$  and  $0.85$ . 10 000 independent samples were obtained for each population, with  $n = 40$  and  $n = 50$  individuals, according to a SRS scheme. These samples were used for calculating estimates of  $P = N^{-1}t_y$  and estimator's variance as well as for providing 95% confidence intervals (based on normality) for  $P$ . Two estimators were considered: one, assisted by a normal regression model and denoted by  $\hat{P}_N$ , and other, assisted by a Bernoulli response model and denoted by  $\hat{P}_B$ . These GREG estimators satisfied the conditions of [Corollary 1](#). Performance measurements were calculated for each estimator as: (i) relative bias of  $\hat{P}$ ; (ii) relative efficiency of  $\hat{P}$  regarding the Horvitz–Thompson estimator; (iii) relative bias of  $\hat{V}(\hat{P})$ ; and (iv) 95% confidence interval coverage rate for  $P$ . [Table 3](#) presents the results. It can be observed that the relative bias of  $\hat{P}_B$  was always smaller than that for  $\hat{P}_N$ ; however, relative bias was very small for both estimators. Regarding relative efficiency, the values were always lower than 100%, indicating that both  $\hat{P}_N$  and  $\hat{P}_B$  estimators were more efficient than the Horvitz–Thompson one. Greater efficiency was always provided for the estimator assisted by the Bernoulli response model. The relative bias of  $\hat{V}(\hat{P})$  was relatively large, mainly for the estimator assisted by the Bernoulli model. The 95% confidence interval coverage rates for  $P$ , for both estimators, were always lower than nominal values; nevertheless, taking into account that the sample size was small, coverage rate values were good. It can be observed that  $\hat{P}_B$  performed better than  $\hat{P}_N$ , mainly when the value of the population proportion  $P$  moves away from 0.5.

### 3.6.2. Poisson response

Nine populations, consisting of  $N = 1000$  individuals each, were generated where one auxiliary variable,  $x$ , was considered, generated by a chi-square distribution with 1 degree of freedom. The  $\hat{\mu}_k^U$  value for element  $k$  from each population was calculated as  $\hat{\mu}_k^U = \exp(\hat{\beta}_{U_1} + \hat{\beta}_{U_2}x_k)$ , and the interest variable,  $y_k$ , was randomly generated following a Poisson distribution with parameter  $\hat{\mu}_k^U$ . The  $\hat{\beta}_U$  values in each population were chosen so that they ensured  $R_U^2 = 0.5, 0.65$  and  $0.75$ , and  $m = N^{-1}t_y = 1.5, 3.0$  and  $5.0$ . 10 000 independent samples were obtained for each population, with sizes  $n = 40$  and  $n = 50$ , according to a SRS scheme. These samples were used for calculating estimates of  $m$  and

**Table 4**  
Performance of  $\hat{m}_N$  and  $\hat{m}_P$  estimating the population mean  $m$ .

| $R_U^2$ | $n$ | $m$ | Measures (%)            |                                      |                                           |       |                         |                                      |                                           |       |
|---------|-----|-----|-------------------------|--------------------------------------|-------------------------------------------|-------|-------------------------|--------------------------------------|-------------------------------------------|-------|
|         |     |     | $\hat{m}_N$             |                                      |                                           |       | $\hat{m}_P$             |                                      |                                           |       |
|         |     |     | $\frac{\hat{m}}{m} - 1$ | $\frac{V(\hat{m})}{V(\hat{m}_{HT})}$ | $\frac{\hat{V}(\hat{m})}{V(\hat{m})} - 1$ | CR    | $\frac{\hat{m}}{m} - 1$ | $\frac{V(\hat{m})}{V(\hat{m}_{HT})}$ | $\frac{\hat{V}(\hat{m})}{V(\hat{m})} - 1$ | CR    |
| 0.50    | 40  | 1.5 | 1.21                    | 60.94                                | 8.37                                      | 91.98 | 0.62                    | 55.58                                | 16.65                                     | 91.25 |
|         |     | 3.0 | 0.78                    | 56.91                                | 8.75                                      | 92.34 | 0.38                    | 55.01                                | 14.71                                     | 91.47 |
|         |     | 5.0 | 0.44                    | 54.13                                | 8.01                                      | 92.83 | 0.19                    | 52.06                                | 11.13                                     | 92.39 |
|         | 50  | 1.5 | 0.98                    | 60.80                                | 6.07                                      | 92.34 | 0.42                    | 55.12                                | 14.48                                     | 91.75 |
|         |     | 3.0 | 0.54                    | 56.36                                | 5.96                                      | 93.04 | 0.30                    | 53.51                                | 10.63                                     | 92.50 |
|         |     | 5.0 | 0.30                    | 55.11                                | 8.49                                      | 92.91 | 0.18                    | 52.66                                | 11.10                                     | 92.58 |
| 0.65    | 40  | 1.5 | 2.00                    | 58.41                                | 14.77                                     | 89.18 | 1.35                    | 49.11                                | 16.42                                     | 88.08 |
|         |     | 3.0 | 1.43                    | 51.69                                | 10.44                                     | 91.05 | 0.37                    | 37.24                                | 15.45                                     | 91.27 |
|         |     | 5.0 | 0.80                    | 45.99                                | 11.18                                     | 91.37 | 0.28                    | 37.23                                | 14.95                                     | 90.99 |
|         | 50  | 1.5 | 1.65                    | 56.64                                | 9.55                                      | 90.50 | 1.01                    | 43.68                                | 17.17                                     | 89.33 |
|         |     | 3.0 | 1.08                    | 51.02                                | 7.72                                      | 91.61 | 0.21                    | 35.85                                | 10.55                                     | 92.26 |
|         |     | 5.0 | 0.76                    | 45.68                                | 9.27                                      | 92.26 | 0.12                    | 36.97                                | 12.76                                     | 92.26 |
| 0.75    | 40  | 1.5 | 3.22                    | 58.22                                | 14.16                                     | 84.41 | 0.65                    | 34.80                                | 14.59                                     | 85.49 |
|         |     | 3.0 | 1.85                    | 49.67                                | 14.06                                     | 88.63 | 0.59                    | 30.20                                | 18.12                                     | 88.74 |
|         |     | 5.0 | 1.17                    | 43.07                                | 12.33                                     | 90.36 | 0.01                    | 26.79                                | 17.01                                     | 91.51 |
|         | 50  | 1.5 | 2.52                    | 58.47                                | 10.11                                     | 86.11 | 0.49                    | 32.03                                | 17.59                                     | 87.22 |
|         |     | 3.0 | 1.41                    | 48.60                                | 10.57                                     | 89.92 | 0.50                    | 28.96                                | 13.10                                     | 89.98 |
|         |     | 5.0 | 1.12                    | 43.02                                | 10.63                                     | 91.02 | 0.16                    | 26.23                                | 13.83                                     | 92.08 |

estimator's variance, as well as 95% confidence intervals (based on normality) for  $m$ . Two estimators were considered: one, assisted by a normal regression model and denoted by  $\hat{m}_N$ , and the other, assisted by a Poisson response model, denoted by  $\hat{m}_P$ . These GREG estimators satisfied the conditions of Corollary 1. The same measurements of performance as in the Bernoulli case were calculated for each estimator. Table 4 presents the results. It can be observed that the relative bias of  $\hat{m}_P$  was always smaller than that for the  $\hat{m}_N$ . However, relative bias was very small for both estimators. Regarding relative efficiency, the values were always lower than 100%, indicating that the  $\hat{m}_N$  and  $\hat{m}_P$  estimators were more efficient than the Horvitz–Thompson one. Once again, greater efficiency always pertained to the estimator assisted by the Poisson model than the one assisted by the normal one. The relative bias of  $\hat{V}(\hat{m})$  was relatively large, mainly for the estimator assisted by the Poisson model. For both estimators, the 95% confidence interval coverage rates for  $m$  were always lower than the nominal values; nevertheless, taking into account that the sample size was small, coverage rate values were good. The  $\hat{m}_P$  performed better than  $\hat{m}_N$ , notoriously when population mean  $m$  value was small.

### 3.6.3. Gamma response

Nine populations consisting of  $N = 1000$  individuals each were generated, where one auxiliary variable,  $x$  was considered, being generated by a chi-square distribution with 1 degree of freedom. The value of  $\hat{\mu}_k^U$  for the element  $k$  for each population was calculated as  $\hat{\mu}_k^U = \left( \hat{\beta}_{U_1} + \hat{\beta}_{U_2} x_k \right)^{-1}$  and the interest variable,  $y_k$ , was randomly generated following a gamma distribution with  $\hat{\mu}_k^U$  and  $\phi$  as parameters. In each population, the values of  $\hat{\beta}_U$  and  $\phi$  were chosen so that  $R_U^2 = 0.5, 0.65$  and  $0.75$ , and  $m = 5, 10$  and  $15$ . 10 000 independent simple random samples were obtained from each population, with sizes  $n = 40$  and  $n = 50$ . These samples were used for calculating estimates of  $m$ , estimator's variance and 95% confidence intervals (based on normality) for  $m$ . Again, two estimators were considered: one assisted by a normal regression model ( $\hat{m}_N$ ) and the other, assisted by a gamma response model ( $\hat{m}_G$ ). These GREG estimators satisfied the conditions of Corollary 1. The same performance measurements as in the Bernoulli and Poisson cases were calculated for each estimator. Table 5 presents the results. It can be observed that the relative bias of  $\hat{m}_G$  was always smaller than that for  $\hat{m}_N$ . However, relative bias was very small for both estimators. Regarding relative efficiency, again the values were always lower than 100%, indicating that the  $\hat{m}_N$  and  $\hat{m}_G$  estimators were more efficient than the Horvitz–Thompson one. Here also the greater efficiency was always given by the estimator assisted by the gamma model. The relative bias of  $\hat{V}(\hat{m})$  was relatively small. For both estimators, 95% confidence interval coverage rates for  $m$  were always lower than nominal values; nevertheless, taking into account that sample size was small, coverage rate values were good. It can be observed that  $\hat{m}_G$  performed better than  $\hat{m}_N$ , mainly when populational mean  $m$  value was small.

## 4. Applications

### 4.1. Academic performance index (API)

The academic performance index (API) has been used annually as a performance measurement for every school in California. The API is based on a standardized test applied to students. Data from 2000 was analyzed. The data set contains

**Table 5**  
Performance of  $\hat{m}_N$  and  $\hat{m}_G$  estimating the population mean  $m$ .

| $R_U^2$ | $n$ | $m$ | Measures (%)              |                                      |                                             |       |                           |                                      |                                             |       |
|---------|-----|-----|---------------------------|--------------------------------------|---------------------------------------------|-------|---------------------------|--------------------------------------|---------------------------------------------|-------|
|         |     |     | $\hat{m}_N$               |                                      |                                             |       | $\hat{m}_G$               |                                      |                                             |       |
|         |     |     | $ \frac{\hat{m}}{m} - 1 $ | $\frac{V(\hat{m})}{V(\hat{m}_{HT})}$ | $ \frac{\hat{V}(\hat{m})}{V(\hat{m})} - 1 $ | CR    | $ \frac{\hat{m}}{m} - 1 $ | $\frac{V(\hat{m})}{V(\hat{m}_{HT})}$ | $ \frac{\hat{V}(\hat{m})}{V(\hat{m})} - 1 $ | CR    |
| 0.50    | 40  | 5   | 1.29                      | 69.75                                | 8.62                                        | 92.02 | 0.19                      | 53.58                                | 4.82                                        | 93.22 |
|         |     | 10  | 0.55                      | 66.39                                | 11.10                                       | 92.04 | 0.03                      | 55.73                                | 5.66                                        | 93.36 |
|         |     | 15  | 0.54                      | 61.45                                | 8.81                                        | 92.20 | 0.09                      | 52.28                                | 4.23                                        | 93.30 |
|         | 50  | 5   | 0.94                      | 67.97                                | 7.62                                        | 92.24 | 0.18                      | 52.76                                | 4.52                                        | 93.24 |
|         |     | 10  | 0.51                      | 63.59                                | 6.45                                        | 93.04 | 0.04                      | 53.27                                | 0.84                                        | 94.34 |
|         |     | 15  | 0.35                      | 58.96                                | 5.37                                        | 93.36 | 0.06                      | 51.15                                | 3.56                                        | 93.84 |
| 0.65    | 40  | 5   | 1.76                      | 72.40                                | 14.26                                       | 90.66 | 0.12                      | 41.65                                | 4.96                                        | 93.08 |
|         |     | 10  | 1.15                      | 58.86                                | 12.61                                       | 91.28 | 0.02                      | 37.61                                | 2.93                                        | 93.96 |
|         |     | 15  | 0.80                      | 53.42                                | 10.63                                       | 92.06 | 0.09                      | 37.60                                | 7.10                                        | 93.32 |
|         | 50  | 5   | 1.38                      | 71.60                                | 12.58                                       | 91.90 | 0.11                      | 41.59                                | 4.67                                        | 93.18 |
|         |     | 10  | 0.87                      | 59.87                                | 12.23                                       | 92.08 | 0.03                      | 37.45                                | 0.81                                        | 94.10 |
|         |     | 15  | 0.54                      | 51.95                                | 10.02                                       | 92.36 | 0.08                      | 37.58                                | 1.75                                        | 93.78 |
| 0.75    | 40  | 5   | 2.19                      | 76.46                                | 12.65                                       | 90.40 | 0.24                      | 34.61                                | 8.66                                        | 92.18 |
|         |     | 10  | 1.47                      | 63.31                                | 18.95                                       | 90.42 | 0.10                      | 32.57                                | 6.74                                        | 92.66 |
|         |     | 15  | 1.23                      | 58.05                                | 19.59                                       | 89.54 | 0.12                      | 29.98                                | 4.12                                        | 93.30 |
|         | 50  | 5   | 1.62                      | 76.49                                | 11.76                                       | 91.26 | 0.16                      | 36.12                                | 6.36                                        | 92.22 |
|         |     | 10  | 1.08                      | 61.48                                | 15.09                                       | 90.76 | 0.10                      | 32.84                                | 6.39                                        | 92.98 |
|         |     | 15  | 0.84                      | 55.31                                | 13.96                                       | 91.06 | 0.06                      | 29.49                                | 4.08                                        | 93.64 |

**Table 6**  
Relative efficiency (%) of  $\hat{m}_G$  regarding to H-T estimator for the API example.

| Sampling scheme | Model assisting | School type |        |             | Total |
|-----------------|-----------------|-------------|--------|-------------|-------|
|                 |                 | Elemental   | Middle | High school |       |
| SRS             | Normal          | 49.11       | 46.48  | 38.76       | 48.95 |
|                 | Gamma           | 44.65       | 40.32  | 33.92       | 43.93 |
| BER             | Normal          | 1.70        | 1.49   | 1.00        | 1.76  |
|                 | Gamma           | 1.64        | 1.41   | 0.95        | 1.59  |

6194 observations and 37 variables and was obtained from the *Survey* package for the statistical software *R* ([www.r-project.org](http://www.r-project.org)). The goal was to estimate the API average for the population of schools using a stratified sample design. The strata were formed by the levels of *Stype* variable (elementary (E), middle (M) and high school (H)). SRS and BER sampling schemes performances were investigated in each strata. The  $\hat{m}_G$  estimator discussed in Section 3.5.6 was applied; the estimation in each stratum was assisted by models with normal and gamma responses, as described in (9) and (13). The percentage of parents not graduated from high school in each school *not.hsg* was the auxiliary variable. Strata sizes were given by  $N_1 = 4421$ ,  $N_2 = 1018$  and  $N_3 = 755$ . Strata sample sizes were 100, 50 and 50, respectively for the first, second and third stratum. Such numbers corresponded to expected sample sizes when using a BER sampling scheme.

Table 6 shows  $\hat{m}_G$  relative efficiency ( $\Delta_{p(\cdot)}$ ) with respect to the Horvitz–Thompson estimator for the two regression models and the two sample schemes. It can be seen that the gamma assisted  $\hat{m}_G$  estimator was more efficient than the normal assisted one in every strata. Normal assisted estimator variance was approximately 1.11 times greater than the variance for the gamma model assisted estimator, for both sampling schemes. Fig. 1 shows the relationship between interest and the auxiliary variable. The  $R_U^2$  values for the normal model's goodness of fit were 0.51, 0.54 and 0.61, for the E, M and H strata, respectively. The  $R_U^2$  values for the gamma model were 0.55, 0.60 and 0.66, for strata E, M and H, respectively.

#### 4.2. Quality of life in Colombia

The unsatisfied basic necessities indicator (UBN) was calculated for the 1118 Colombian municipalities, based on the household and population census carried out by the National Administrative Department of Statistics–DANE ([www.dane.gov.co](http://www.dane.gov.co)). The data set was related to information from 2005. The UBN is an indicator of whether the population's basic needs are been provided for, concerning (i) inadequacy housing, (ii) overpopulated households, (iii) households having inadequate services, (iv) high economic dependence households and (v) families with children that do not attend school. The goal was to estimate the proportion of municipalities (excluding the capital Bogota) presenting a higher than 25% UBN. A stratified sample design was used with strata based on a *Distance* variable defined as the distance, in kilometers, between each municipality and the closest city having more than 1 million inhabitants. Three strata were considered: one for distances less than 150 km (<150 km), one for distances between 150 and 200 km (150–200 km), and another for

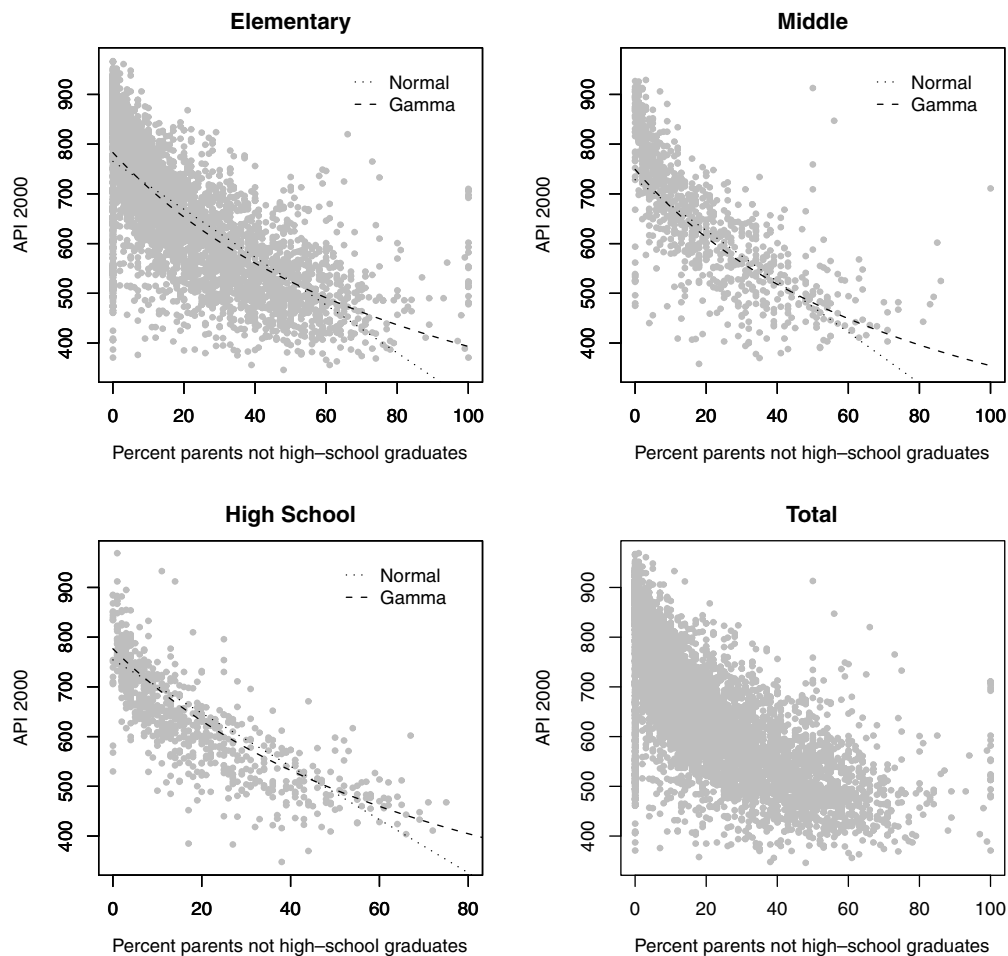


Fig. 1. Scatter plots between interest and auxiliary variable for the API example.

Table 7

Relative efficiency (%) of  $\hat{m}_G$  regarding to H-T estimator for the *Quality of life in Colombia* application when the interest variable is dichotomous.

| Sampling scheme | Model assisting | Distance (km) |         |       | Total |
|-----------------|-----------------|---------------|---------|-------|-------|
|                 |                 | <150          | 150–200 | >200  |       |
| SRS             | Normal          | 93.09         | 72.76   | 87.43 | 85.61 |
|                 | Bernoulli       | 76.60         | 59.82   | 80.61 | 72.41 |
| BER             | Normal          | 21.68         | 6.99    | 6.60  | 14.93 |
|                 | Bernoulli       | 21.25         | 6.86    | 6.43  | 12.57 |

distances over 200 km ( $>200$  km). SRS and BER schemes performances were investigated in each strata. The  $\hat{m}_G$  estimator, discussed in the Section 3.5.6, was applied. The estimation in each strata was assisted by models with normal and Bernoulli responses, as described in (9) and (11). The variable *Density*, the average number of people living in an area of 1 km<sup>2</sup> of the city was the auxiliary variable. Stratum sizes were  $N_1 = 308$ ,  $N_2 = 395$  and  $N_3 = 414$ , respectively. Stratum sample sizes were 46, 59 and 62, respectively, for the first, second and third stratum. These sizes were expected values when using BER sampling.

Table 7 shows the relative efficiency ( $\Delta_{p(\cdot)}$ ) of the  $\hat{m}_G$  estimator compared to the Horvitz–Thompson estimator, for both regression models and sample schemes. It can be seen that the Bernoulli assisted  $\hat{m}_G$  estimator was more efficient than the normal assisted one in every strata. The variance of the normal assisted estimator was approximately 1.18 times greater than the variance of the Bernoulli model assisted estimator for both sampling schemes. Fig. 2 shows the relationship between interest and auxiliary variables. The  $R^2_U$  values for the normal model's goodness of fit were 0.06, 0.27 and 0.13, for the first

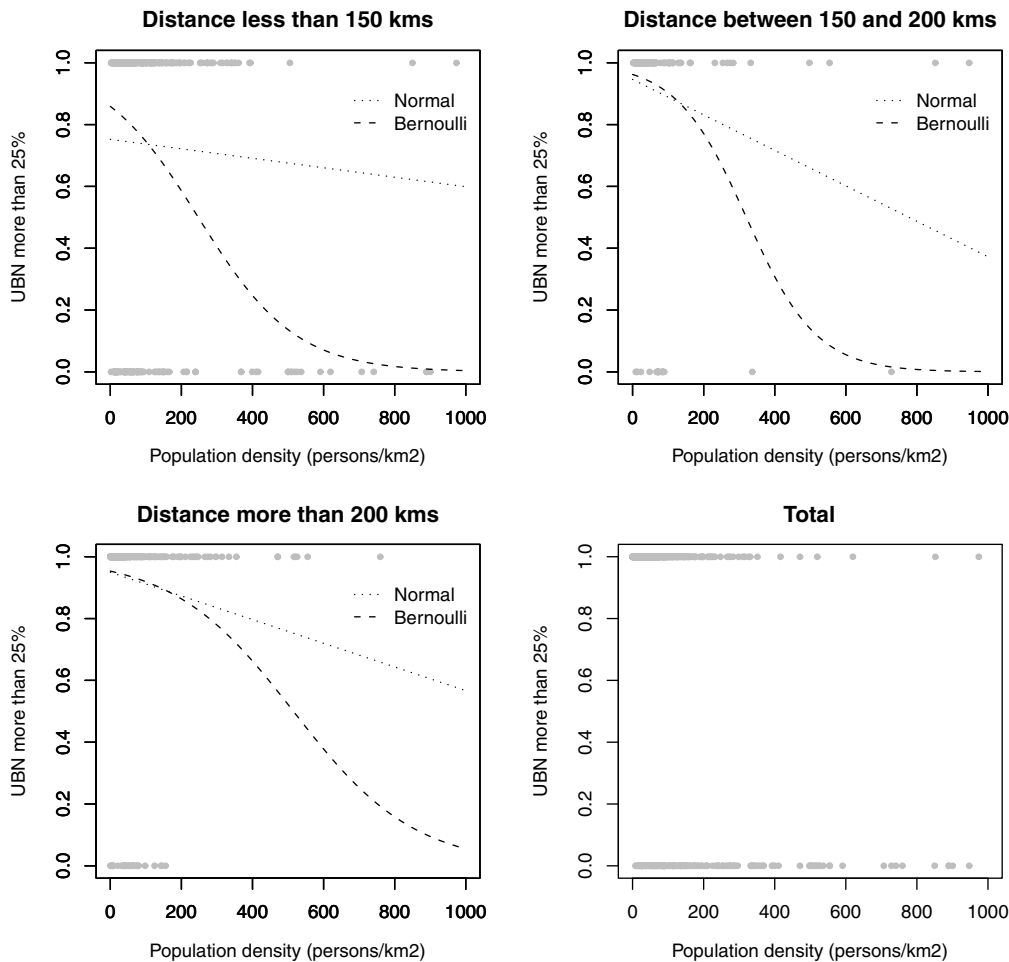


Fig. 2. Scatter plots between interest and auxiliary variables for the *Quality of life in Colombia* application when the interest variable is dichotomous.

**Table 8**

Relative efficiency (%) of  $\hat{m}_G$  with respect to the H–T estimator for the *Quality of life in Colombia* application when the interest variable is a counting process.

| Sampling scheme | Model assisting | Distance (km) |         |       | Total |
|-----------------|-----------------|---------------|---------|-------|-------|
|                 |                 | <150          | 150–200 | >200  |       |
| SRS             | Normal          | 93.88         | 83.84   | 90.37 | 89.14 |
|                 | Poisson         | 76.42         | 73.23   | 79.09 | 76.23 |
| BER             | Normal          | 23.22         | 13.01   | 11.16 | 17.68 |
|                 | Poisson         | 22.21         | 12.66   | 10.77 | 14.91 |

(<150 km), the second (150–200 km) and the third (>200 km) strata, respectively. The  $R_U^2$  values for the Bernoulli model were 0.24, 0.40 and 0.20, respectively.

Additionally, we wanted to estimate the average number of UBN components higher than 10%. In this case, possible interest variable values were 0, 1, 2, 3, 4 and 5, thereby motivating an estimator assisted by a Poisson regression model. Table 8 shows the relative efficiency ( $\Delta_{p(\cdot)}$ ) of the  $\hat{m}_G$  estimator compared to the Horvitz–Thompson estimator for both regression models and sample schemes. It can be seen that the Poisson assisted  $\hat{m}_G$  estimator was more efficient than the normal assisted one in all strata. Normal assisted estimator variance was about 1.17 times greater than the variance for the Poisson model assisted estimator, for both sampling schemes. Fig. 3 shows the relationship between interest and auxiliary variables. The  $R_U^2$  values for the normal model's goodness of fit were 0.06, 0.16 and 0.10 for the strata <150 km, 150–200 km and >200 km, respectively. The  $R_U^2$  values for the Poisson model were 0.24, 0.27 and 0.21, for the <150 km, 150–200 km and >200 km strata, respectively.

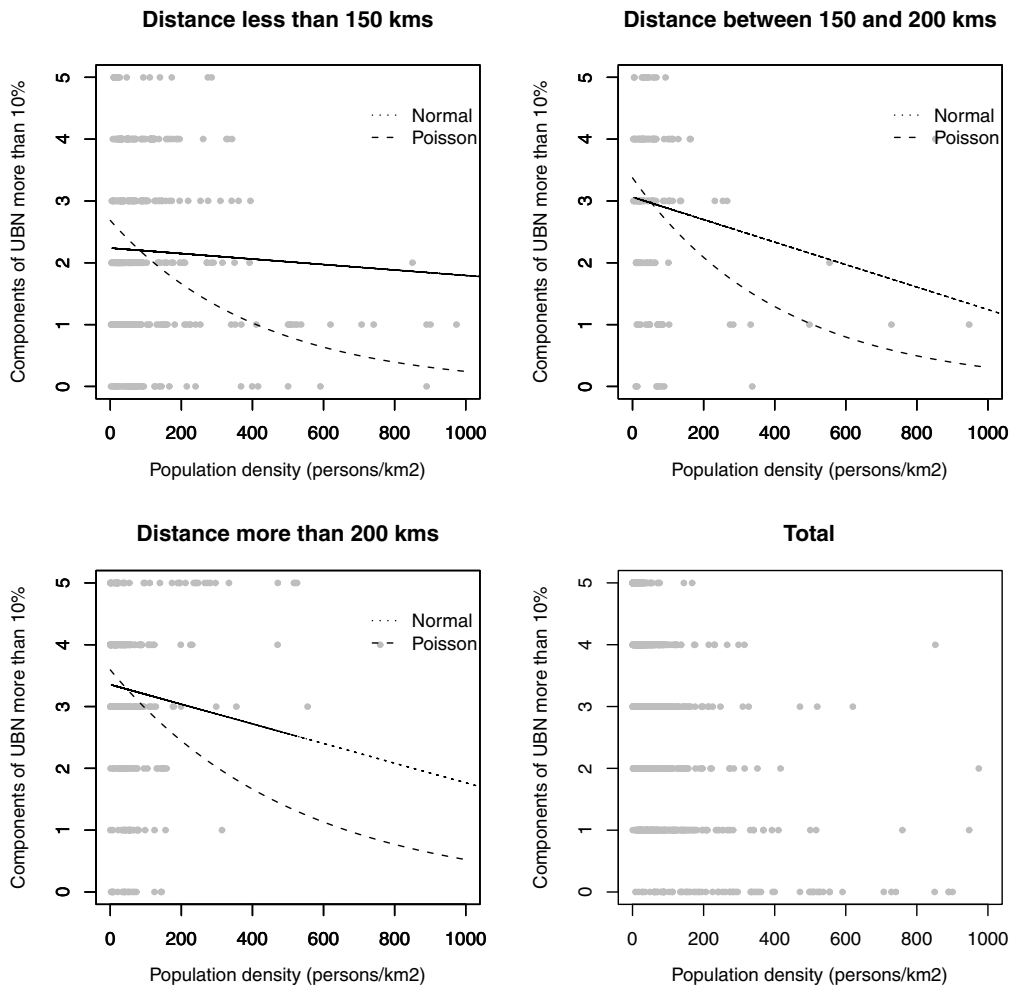


Fig. 3. Scatter plots between interest and auxiliary variables for the *Quality of life in Colombia* example when the interest variable is a counting process.

## Acknowledgments

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## Appendix A. Theorem 1

### Proof.

- (i)  $\hat{\beta}_U$  estimate satisfies  $\mathbf{U}_U(\hat{\beta}_U) = \mathbf{0}$ , i.e.,  $U_{U_j}(\hat{\beta}_U) = \mathbf{h}_j^T \mathbf{E} = 0, j = 1, \dots, J$ , where  $\mathbf{h}_j = (h_{kj}, \dots, h_{Nj})^T$  and  $\mathbf{E} = (E_1, \dots, E_N)^T$ , with  $h_{kj} = \phi_k \frac{\partial \theta_k}{\partial \eta_k} x_{kj}$ . If exist  $\mathbf{a} = (a_1, \dots, a_J)^T$  such that  $\sum_{j=1}^J a_j \mathbf{h}_j = \mathbf{1}_N$ , then,  $\sum_{k \in U} (y_k - \hat{\mu}_k^U) = \mathbf{1}_N^T \mathbf{E} = \left( \sum_{j=1}^J a_j \mathbf{h}_j \right)^T \mathbf{E} = \sum_{j=1}^J a_j \mathbf{h}_j^T \mathbf{E}$ . From  $U_{U_j}(\hat{\beta}_U) = 0$  above,  $j = 1, \dots, J$ , we have  $\sum_{j=1}^J a_j \mathbf{h}_j^T \mathbf{E} = 0$ , therefore,  $t_y = \sum_{k \in U} \hat{\mu}_k^U$ .
- (ii)  $\hat{\beta}_S^\pi$  estimate satisfies  $\mathbf{U}_S^\pi(\hat{\beta}_S^\pi) = \mathbf{0}$ , i.e.,  $U_{S_j}^\pi(\hat{\beta}_S^\pi) = \mathbf{h}_{S_j}^T \mathbf{e}^* = 0, j = 1, \dots, J$ , where  $\mathbf{h}_{S_j}$  and  $\mathbf{e}^*$  are  $n$ -dimension column vectors with elements  $h_{kj}$  and  $e_k/\pi_k$ , respectively, with  $k \in S$ . If there exists  $\mathbf{a}$  such that  $\sum_{j=1}^J a_j \mathbf{h}_j = \mathbf{1}_N$ , then,  $\sum_{j=1}^J a_j \mathbf{h}_{S_j} = \mathbf{1}_n$ . Therefore,  $\sum_{k \in S} \frac{(y_k - \hat{\mu}_k^S)}{\pi_k} = \mathbf{1}_n^T \mathbf{e}^* = \left( \sum_{j=1}^J a_j \mathbf{h}_{S_j} \right)^T \mathbf{e}^* = \sum_{j=1}^J a_j \mathbf{h}_{S_j}^T \mathbf{e}^*$ . From  $U_{S_j}^\pi(\hat{\beta}_S^\pi) = 0$  above,  $j = 1, \dots, J$ , we have  $\sum_{j=1}^J a_j \mathbf{h}_{S_j}^T \mathbf{e}^* = 0$ , then,  $\hat{t}_G = \sum_{k \in U} \hat{\mu}_k^S$ .  $\square$



## Appendix B. Corollary 1

**Proof.** Homogeneity in the dispersion parameter, systematic component with intercept and canonical link led to  $h_{kj} = \phi x_{kj}$ , where  $h_{k1} = \phi$ . Then, with  $a_1 = \phi^{-1}$  and  $a_j = 0$  for  $j \neq 1$ , we have  $\sum_{j=1}^J a_j \mathbf{h}_j = \mathbf{1}_n$ . Therefore, using Theorem 1 we could prove that  $t_y = \sum_{k \in U} \hat{\mu}_k^U$  and  $\hat{t}_G = \sum_{k \in U} \hat{\mu}_k^S$ .  $\square$

## Appendix C. Corollary 2

**Proof.** Normal response, systematic component with canonical link and dispersion parameter expressed as  $\phi_k^{-1} = \sum_{j=1}^J a_j x_{kj}$  led to  $h_{kj} = \frac{x_{kj}}{\sum_{r=1}^J a_r x_{kr}}$ . We could verify that  $\sum_{j=1}^J a_j \mathbf{h}_j = \mathbf{1}_N$ . Therefore, using Theorem 1 we could prove that  $t_y = \sum_{k \in U} \hat{\mu}_k^U$  and  $\hat{t}_G = \sum_{k \in U} \hat{\mu}_k^S$ .  $\square$

**Lemma 1.** If the  $\hat{t}_G$  is based on a  $\xi$  model where it has been verified that there is homogeneity in the dispersion parameter, systematic component with intercept and canonical link, then  $\hat{\boldsymbol{\mu}}^T(\mathbf{Y}_U - \hat{\boldsymbol{\mu}}) \approx \mathbf{0}$  with  $\hat{\boldsymbol{\mu}} = (\hat{\mu}_1^U, \dots, \hat{\mu}_N^U)^T$ .

**Proof.** Expanding  $\hat{\mu}_k^U$  by Taylor series of first-order around of  $\mathbf{x}_k^\circ = (1, x_{k2}^\circ, \dots, x_{kJ}^\circ)^T$  we have the following

$$\hat{\mu}_k^U = \mu(\mathbf{x}_k, \hat{\boldsymbol{\beta}}_U) \approx \hat{\mu}_k^\circ + \sum_{j=2}^J V(\hat{\mu}_k^\circ) \hat{\beta}_{Uj} (x_{kj} - x_{kj}^\circ),$$

where  $\hat{\mu}_k^\circ = \mu(\mathbf{x}_k^\circ, \hat{\boldsymbol{\beta}}_U)$ . For example, with  $x_{kj}^\circ = N^{-1} t_{x_j} = \bar{x}_{Uj}$ ,  $j = 2, \dots, J$ , we have that  $\mathbf{x}_k^\circ = \mathbf{x}^\circ$  and therefore  $\hat{\mu}_k^\circ = \hat{\mu}^\circ$ . Hence

$$\begin{aligned} \hat{\mu}_k^U &\approx \hat{\mu}^\circ + \sum_{j=2}^J V(\hat{\mu}^\circ) \hat{\beta}_{Uj} (x_{kj} - \bar{x}_{Uj}) \\ &= \sum_{j=1}^J m_j x_{kj}, \end{aligned}$$

where  $m_j = V(\hat{\mu}^\circ) \hat{\beta}_{Uj}$  for  $j \neq 1$  and  $m_1 = \hat{\mu}^\circ - \sum_{j=2}^J V(\hat{\mu}^\circ) \hat{\beta}_{Uj} \bar{x}_{Uj}$ . In matrix form  $\hat{\boldsymbol{\mu}}$  can be written as

$$\hat{\boldsymbol{\mu}} \approx \mathbf{X}_U \mathbf{m},$$

with  $\mathbf{m} = (m_1, \dots, m_J)^T$ .  $\hat{\boldsymbol{\beta}}_U$  satisfies  $\mathbf{U}_U(\hat{\boldsymbol{\beta}}_U) = \mathbf{0}$ . As  $\hat{t}_G$  is based on a  $\xi$  model where it has been verified that there is homogeneity in the dispersion parameter, systematic component with intercept and canonical link,  $\mathbf{U}_U(\boldsymbol{\beta})$  can be expressed as  $\mathbf{U}_U(\boldsymbol{\beta}) = \mathbf{X}_U^T(\mathbf{Y}_U - \boldsymbol{\mu})$ . Therefore, we have  $\mathbf{X}_U^T(\mathbf{Y}_U - \hat{\boldsymbol{\mu}}) = \mathbf{0}$ , and

$$\begin{aligned} \hat{\boldsymbol{\mu}}^T(\mathbf{Y}_U - \hat{\boldsymbol{\mu}}) &\approx (\mathbf{X}_U \mathbf{m})^T (\mathbf{Y}_U - \hat{\boldsymbol{\mu}}) \\ &= \mathbf{m}^T \mathbf{X}_U^T (\mathbf{Y}_U - \hat{\boldsymbol{\mu}}) \\ &= \mathbf{m}^T \mathbf{0} \\ &= 0. \quad \square \end{aligned}$$

## Appendix D. Theorem 2

**Proof.** According to Pregibon (1981), in the convergence of the iterative process to solve  $\mathbf{U}_S^\pi(\hat{\boldsymbol{\beta}}_S^\pi) = \mathbf{0}$ ,  $\hat{\boldsymbol{\beta}}_S^\pi$  can be expressed as

$$\begin{aligned} \hat{\boldsymbol{\beta}}_S^\pi &= (\mathbf{X}_S^T \hat{\mathbf{V}}_S^* \mathbf{X}_S)^{-1} \mathbf{X}_S^T \hat{\mathbf{V}}_S^* \hat{\mathbf{Y}}_S^T \\ &\approx (\mathbf{X}_S^T \hat{\mathbf{V}}_S^* \mathbf{X}_S)^{-1} \mathbf{X}_S^T \hat{\mathbf{V}}_U^* \hat{\mathbf{Y}}_U^T, \end{aligned} \quad (\text{D.1})$$

where  $\hat{\mathbf{V}}_S^* = \text{diag}\{\hat{V}_1^S \phi_1 / \pi_1, \dots, \hat{V}_n^S \phi_n / \pi_n\}$ ,  $\hat{\mathbf{V}}_U^* = \text{diag}\{\hat{V}_1^U \phi_1 / \pi_1, \dots, \hat{V}_n^U \phi_n / \pi_n\}$ ,  $\hat{\mathbf{Y}}_S^* = (\hat{y}_1^S, \dots, \hat{y}_n^S)^T$ ,  $\hat{\mathbf{Y}}_U^* = (\hat{y}_1^U, \dots, \hat{y}_n^U)^T$  with  $\hat{y}_k^S = \{\eta_k + V_k^{-1}(y_k - \mu_k)\}_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}_S^\pi}$  and  $\hat{y}_k^U = \{\eta_k + V_k^{-1}(y_k - \mu_k)\}_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}_U^\pi}$ . From result 5.10.1 of Särndal et al. (2003) we have that  $\hat{\boldsymbol{\beta}}_S^\pi$  (as in the expression (D.1)) is an approximately unbiased estimator of  $\hat{\boldsymbol{\beta}}_U$  (i.e.  $E(\hat{\boldsymbol{\beta}}_S^\pi - \hat{\boldsymbol{\beta}}_U) \approx \mathbf{0}$ ) with the approximate variance-covariance matrix given by

$$(\mathbf{X}_U^T \mathbf{W}_U^* \mathbf{X}_U)^{-1} \Sigma_U(\mathbf{X}_U^T \mathbf{W}_U^* \mathbf{X}_U)^{-1}.$$

Expanding the GREG estimator  $\hat{t}_G$  by Taylor series of first-order around of  $\hat{\beta}_U$  we have the following

$$\begin{aligned}\hat{t}_G &\approx \sum_{k \in U} \hat{\mu}_k^U + \sum_{j=1}^J \sum_{k \in U} \hat{V}_k^U x_{kj} (\hat{\beta}_{S_j}^\pi - \hat{\beta}_{U_j}) \\ &\approx t_y + \mathbf{1}_N^T \hat{\mathbf{V}}_U \mathbf{X}_U (\hat{\beta}_S^\pi - \hat{\beta}_U).\end{aligned}$$

Then, we can write

- (i)  $E(\hat{t}_G) \approx t_y + \mathbf{1}_N^T \hat{\mathbf{V}}_U \mathbf{X}_U E(\hat{\beta}_S^\pi - \hat{\beta}_U) = t_y$ ;
- (ii)  $\text{Var}(\hat{t}_G) \approx \mathbf{1}_N^T \hat{\mathbf{V}}_U \mathbf{X}_U \text{Var}(\hat{\beta}_S^\pi - \hat{\beta}_U) \mathbf{X}_U^T \hat{\mathbf{V}}_U \mathbf{1}_N = \Gamma_U^T \Sigma_U \Gamma_U$ ,  
and the  $\text{Var}(\hat{t}_G)$  estimator can be obtained replacing the unknown quantities by its estimators in  $\text{Var}(\hat{t}_G)$ , i.e.,  $\hat{V}_k^U$  by  $\hat{V}_k^S$  and  $\sigma_{ij}^U$  by  $\sigma_{ij}^S$ , as well;
- (iii)  $\hat{\text{Var}}(\hat{t}_G) = \Gamma_S^T \Sigma_S \Gamma_S$ .  $\square$

### Appendix E. Corollary 3

**Proof.** As  $\hat{t}_G$  is based on a  $\xi$  model where it has been verified that there is homogeneity in the dispersion parameter and systematic component with intercept and canonical link we have

$$\Gamma_U = (\mathbf{X}_U^T \hat{\mathbf{V}}_U^* \mathbf{X}_U)^{-1} \mathbf{X}_U^T \hat{\mathbf{V}}_U^* \mathbf{1}_N = (1, 0, \dots, 0)_{(J \times 1)}^T.$$

Then,  $\text{Var}(\hat{t}_G)$  is the (1, 1) element of  $\Gamma_U$ , i.e.,  $\text{Var}(\hat{t}_G) \approx \sigma_{11}^U$  and  $\hat{\text{Var}}(\hat{t}_G) \approx \sigma_{11}^S$ , where  $x_{k1} = \phi_k = 1$  for  $k \in U$ .  $\square$

### Appendix F. Theorem 3

**Proof.** Using Pearson's linear correlation coefficient definition we can write

$$1 - R_U^2 = 1 - \frac{\left[ \sum_{k \in U} y_k \hat{\mu}_k^U - N \bar{y}_U \bar{\hat{\mu}}_U \right]^2}{\left( \sum_{k \in U} y_k^2 - N \bar{y}_U^2 \right) \left( \sum_{k \in U} (\hat{\mu}_k^U)^2 - N \bar{\hat{\mu}}_U^2 \right)},$$

where  $\bar{\hat{\mu}}_U = N^{-1} \sum_{k \in U} \hat{\mu}_k^U$ . As the conditions of Corollary 1 are satisfied we have  $\sum_{k \in U} (y_k - \hat{\mu}_k^U) = 0$ . Then,  $\bar{y}_U = \bar{\hat{\mu}}_U$  and we can write the following

$$1 - R_U^2 = 1 - \frac{\left[ \sum_{k \in U} y_k \hat{\mu}_k^U - N \bar{y}_U^2 \right]^2}{\left( \sum_{k \in U} y_k^2 - N \bar{y}_U^2 \right) \left( \sum_{k \in U} (\hat{\mu}_k^U)^2 - N \bar{y}_U^2 \right)}$$

From Lemma 1 we have  $\hat{\mu}^T (\mathbf{Y}_U - \hat{\mu}) \approx 0$ , i.e.,  $\sum_{k \in U} y_k \hat{\mu}_k^U \approx \sum_{k \in U} (\hat{\mu}_k^U)^2$ , therefore

$$\begin{aligned}1 - R_U^2 &\approx 1 - \frac{\left[ \sum_{k \in U} \hat{\mu}_k^{U2} - N \bar{y}_U^2 \right]^2}{\left( \sum_{k \in U} y_k^2 - N \bar{y}_U^2 \right) \left( \sum_{k \in U} (\hat{\mu}_k^U)^2 - N \bar{y}_U^2 \right)} \\ &= 1 - \frac{\sum_{k \in U} (\hat{\mu}_k^U)^2 - N \bar{y}_U^2}{\sum_{k \in U} y_k^2 - N \bar{y}_U^2} \\ &= \frac{\sum_{k \in U} y_k^2 - 2 \sum_{k \in U} y_k \hat{\mu}_k^U + \sum_{k \in U} \hat{\mu}_k^{U2}}{\sum_{k \in U} y_k^2 - N \bar{y}_U^2} \\ &= \frac{\sum_{k \in U} (y_k - \hat{\mu}_k^U)^2}{\sum_{k \in U} (y_k - \bar{y}_U)^2} = \Delta_{AAS}. \quad \square\end{aligned}$$

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